

In addition to the online homework, please hand in the following problems:

Problem 1: Let $\mathbf{F} = [x, y, z]$. Find the flux of $\mathbf{F}$ through each of the surfaces below.

- a. $S : z = 4 - x^2 - y^2$ lying above the xy -plane and oriented by an upward pointing normal.
- b. $S : 4x + 8y + z = 4$ in the first octant with an upward pointing normal.
- c. $S : x^2 + y^2 + z^2 = 4$ with outward pointing normal.